

The Effects of Low pH and CO₂ on Bovine Corneal Lactate Production

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Purpose: Hypoxic corneal swelling in human subjects is reduced by CO₂ induced corneal acidosis. We investigated whether this effect could be explained by a reduction in lactate production under acidic conditions.

Methods: Lactate production from isolated bovine corneas was assayed under normoxic and hypoxic conditions at normal pH (7.5) and low pH (7.05). Low pH was achieved using a CO₂/HCO₃⁻ Ringer or a HEPES buffered Ringer.

Results: Normoxic corneas produced significantly less (~25%) lactate under all acidic conditions. In contrast, hypoxic corneas produced the same or slightly less lactate under acidic conditions (weighted average of all conditions = 6% less lactate).

Conclusions: From these results, we concluded that pH can have a substantial effect on lactate production, but its influence on glycolytic activity during hypoxia is suppressed and may be negligible.

Introduction

Contact lens associated corneal hypoxia can cause corneal swelling by an osmotic process secondary to the accumulation of lactate.^{1,2} In addition to corneal edema, hypoxia increases the production of protons via increased glycolytic activity and depression of oxidative pathways. Thus, anterior surface hypoxia alone has been shown to significantly lower epithelial³ and stromal pH,⁴ but not endothelial pH.⁵ Lens wear can also limit the escape of CO₂ from the cornea, which further decreases the pH of epithelial cells³ and stroma⁴ and also lowers endothelial pH.⁵ Although corneal acidosis has been shown to alter endothelial morphology^{6,7} and function,⁸ the full clinical implications are not known.

Corneal swelling induced by very low oxygen transmissibility with rigid or hydrogel contact lenses is greater than the swelling induced by pure anoxia delivered by 100% nitrogen gas flowing through tight fitting goggles. McNamara and coworkers⁹ considered that this difference in swelling could be due to the lower corneal pH known to occur in the lens wearing condition. Paradoxically, eyes exposed to 100% nitrogen gas swelled more than paired eyes exposed to 95% nitrogen:5% CO₂,⁹ a condition that produces significantly greater acidosis than hypoxia alone.^{3,4} Two possible mechanisms were offered for this observed reduction in human corneal swelling under low pH conditions:⁹ 1) a

reduction in the rate of cellular lactate production; and/or 2) reduced hydraulic conductivity of the corneal endothelium, which would slow the rate of fluid influx for a given osmotic load.

Glycolytic activity is suppressed by decreased pH primarily by inhibition of phosphofructokinase activity.¹⁰ Therefore, lactate production and corneal swelling may be reduced during corneal acidosis. In the current study, we asked whether corneal lactate production would be lowered by CO₂ induced acidosis under normoxic and hypoxic conditions. We confirmed previous studies,^{1,2,11} showing that hypoxia alone significantly increases corneal lactate production. However, significant reductions in lactate production under acidic conditions were observed during normoxia, while the results were equivocal during hypoxia.

Materials and methods

Corneas: Bovine eyes free of any gross corneal abnormality were obtained from a local slaughterhouse. The eyes were rinsed with saline and kept on ice during transport to the lab. Approximately 3–4 hours elapsed between death and experimental use. A 17.5 mm full thickness corneal button free of limbal tissue was excised using a steel trephine. Each corneal button was rinsed three times with a freshly prepared Ringer

solution corresponding to the experimental condition as outlined below and left in this solution at room temperature briefly (5 minutes) until the experiment began.

Ringer Solutions: Two Ringer solutions were utilized. A bicarbonate-free Ringer solution was composed of 115 mM NaCl, 2 mM KCl, 1 mM K_2HPO_4 , 28.5 mM Na-Gluconate, 10 mM HEPES, 0.61 mM $MgCl_2$, 1.4 mM Ca-Gluconate, and 5.2 mM glucose. The solution was warmed to 37°C and the pH was adjusted to 7.05 or 7.50 with 1 N NaOH. A bicarbonate Ringer contained the same composition as the bicarbonate free solution except 10 mM of the Na-Gluconate was substituted with 10 mM $NaHCO_3$. The bicarbonate Ringer was warmed to 37°C and bubbled with 5% CO_2 /95% Air for 20 minutes. The pH was adjusted to 7.05 using 1 N NaOH. The osmolarity of both Ringer solutions was adjusted to 300 mOSM with sucrose.

Experimental Protocol: Lactate was assayed from Ringer solutions bathing corneal buttons that were assigned to one of four test conditions: (1) HCO_3^- -free Ringer equilibrated with 100% air, pH 7.5; (2) HCO_3^- -free Ringer equilibrated with 100% N_2 , pH 7.5; (3) 10 mM HCO_3^- Ringer equilibrated with 5% CO_2 /95% air, pH 7.05; and, (4) 10 mM HCO_3^- Ringer equilibrated with 5% CO_2 /95% N_2 , pH 7.05. The 5% CO_2 , pH 7.05 was chosen since this was the condition during corneal exposure to 5% CO_2 /95% N_2 delivered through goggles.⁹ All four conditions were run simultaneously. The Ringer solutions were bubbled with the appropriate gas mixtures for 10 minutes prior to receiving a corneal button. The cap of a glass scintillation vial was drilled with three 1 mm holes to allow for inlet and outlet of the gas and to insert a 20-gauge, 2-inch needle for solution sampling. The bathing solutions were sampled (100 μ L with replacement) every 10 minutes for 60 minutes. All samples were frozen until lactate assays were performed on a separate day. Corneas were removed, dried for 1 week at 50°C, and weighed.

Lactate Assays: Lactate concentration was determined spectrophotometrically (340 nm) by measuring, in duplicate, the generation of reduced nicotinamide adenine dinucleotide (NADH) in the presence of lactate dehydrogenase and excess NAD (Sigma kit #826). For the rate of lactate production, 5% of the previous sample concentration was added to correct for each 100 μ L sample volume withdrawn. Lactate production was expressed as mM/gram dry tissue/hr.

Statistics: The rates of lactate concentration change during the 60 minute test period were calculated by linear regression and expressed as mM per gram of dry tissue per hour. To test whether acidic conditions lowered lactate production rates, comparisons of the means were performed between normoxia and acidic normoxia, as well as between hypoxia and acidic hypoxia using standard one-sided t-test analyses at $P = 0.05$.

Results

The first experimental series tested the rate of lactate production using a total of 32 corneal buttons (eight corneas in each of the four test conditions) and sampling every 10 minutes for 1 hour. The average rates of lactate production were determined by using a linear least squares fit to find the slope. The rate of change in lactate concentration sampled from the bathing

solutions was linear for all four conditions over 60 minutes. Table I (trial #1) summarizes these results for each condition expressed as the mean production after 1 hour. The acidic normoxic condition showed a significant decrease (33%, $P < 0.005$) in the rate of lactate production versus the normoxic condition. Similarly, lactate production in acidic hypoxia was 15% lower than in the hypoxia alone, and was also significant, but only at $P = 0.03$.

In an attempt to show more robust differences in lactate production between acidic hypoxia and nonacidic hypoxia, the experiments were repeated under conditions that we thought would reduce sampling variability. Constant gas circulation leads to small amounts of evaporation from the vial (data not shown). This evaporative rate is dependent on the rate of gas circulation. Although we attempted to equalize gas flow to the vials, small differences were unavoidable and this could have contributed to the variability in lactate concentration when the sample was taken. Furthermore, since lactate efflux is linear over 60 minutes, it was not necessary to sample every 10 minutes continuously. Therefore, we repeated the experiment by first equilibrating the Ringer with the appropriate gas mixture for 10 minutes at 37°C. Each Ringer (3 mL) was then placed in the vials and a cornea was added. After placing the cornea in each vial, the Ringers were bubbled with gases for an additional 5 minutes before the inlet and outlet tubes to each vial were clamped. All the vials were incubated in a shaking water bath at 37°C for an additional 55 minutes. A 1 mL sample of the bathing solutions was taken at the end of 60 minutes. The closed system thereby minimized evaporation of Ringer and also eliminated possible errors due to continuous sampling with replacement. The solution pH at the end of 60 minutes incubation without a cornea was the same as the solution pH at the beginning, indicating that the protocol maintained the differences in incubation among the four conditions.

Table I (trial #2) shows the amount of lactate produced when only one sample was taken at the end of 1 hour. Lactate levels were higher than in trial #1 and variability was somewhat higher. Acidosis significantly slowed lactate production under normoxia by 15% ($P = 0.045$). However, the differences in lactate production between anoxia and acidic anoxia seen previously were not observed in this trial series.

In the experiments above, CO_2 was used in the acidic condition in order to simulate the *in vivo* nitrogen/ CO_2 goggle experiments.⁹ To test whether the above results were due to pH differences or the presence of CO_2 , we repeated the experiments in the total absence of CO_2/HCO_3^- . The Ringers used were bicarbonate-free, buffered with 10 mM HEPES, and corneas were assigned one of four conditions: 1) 100% air, pH 7.5; 2) 100% N_2 , pH 7.5; 3) 100% air, pH 7.05; and, 4) 100% N_2 , pH 7.05. The pH was adjusted with NaOH. Each vial was equilibrated with the appropriate gases for 10 minutes prior to receiving a cornea. Each vial was bubbled with gases for an additional 5 minutes, the inlet and outlet tubes were clamped, and the vials were then incubated in a shaking water bath at 37°C for an additional 55 minutes. A 1 mL sample of the bathing solution was taken at the end of the 60 minutes and assayed for lactate.

TABLE I Lactate production (mM lactate/gm dry tissue $\times$ hr $\pm$ S.D.) by the bovine cornea under normoxic, hypoxic, acidic normoxic, and acidic hypoxic conditions

Trial number	Normoxia*	Acidic normoxia†	Hypoxia‡	Acidic hypoxia§
1 (sampling every 10 minutes)	9.21 $\pm$ 2.10 (8)	6.58 $\pm$ 1.35 (8) [¶] ; -29%	13.03 $\pm$ 2.14 (8)	11.06 $\pm$ 1.58 (8) [¶] ; -15%
2 (sample at 60 minutes only)	9.57 $\pm$ 2.08 (12)	8.11 $\pm$ 2.00 (12) [¶] ; -15%	13.07 $\pm$ 2.62 (12)	13.25 $\pm$ 2.28 (12); +2%
3 (no CO ₂)	10.59 $\pm$ 0.71 (6)	7.91 $\pm$ 1.41 (6) [¶] ; -25%	13.12 $\pm$ 2.02 (6)	11.90 $\pm$ 1.16 (6); -9%
4 (pre-incubate 60 minutes)	4.14 $\pm$ 0.96 (5)	3.50 $\pm$ 0.54 (5) [¶] ; -26%	7.45 $\pm$ 1.07 (5)	8.24 $\pm$ 1.13 (5); +10%
5 (no endothelium, no CO ₂)	8.67 $\pm$ 2.26 (8)	5.80 $\pm$ 1.51 (8) [¶] ; -33%	12.03 $\pm$ 2.20 (8)	10.01 $\pm$ 1.70 (8); [#] -17%

N indicated in parentheses.

* (100% air, pH 7.5).

† (5% CO₂/95% air or 100% air, pH 7.05).

‡ (100% N₂, pH 7.5).

§ (5% CO₂/95% N₂ or 100% N₂, pH 7.05).

¶ Acidic normoxia significantly less than normoxia ($P < 0.05$).

Acidic hypoxia significantly less than hypoxia ($P < 0.05$).

Table I (trial #3) shows that in the absence of CO₂ (low pH), lactate production decreased significantly by 25% ($P < 0.005$) in the acidic versus the nonacidic normoxic condition. In the hypoxic conditions, acidosis in the absence of CO₂ decreased lactate production 9% over the nonacidic condition. However, this small difference was not significant ($P > 0.05$).

Previous studies¹¹ have shown that initial lactate efflux from fresh rabbit corneas may be due to simple diffusion of extracellular lactate over the first 60 minutes. Thereafter, lactate efflux slows considerably, but can be stimulated by exogenous glucose. To determine if our results could be influenced by this initial spontaneous efflux, we preincubated excised corneas in 10 mL of glucose free Ringer at room temperature for 60 minutes. Sterile techniques were employed and the Ringer was continuously agitated using a clinical rotary table. The corneas then underwent the same procedures and four conditions as in the first experiment.

Table I (trial #4) shows that preincubation in 0 glucose caused a reduction in total lactate production. However, the relative comparison between the different conditions was similar to the former experiments where the corneas were immediately subjected to the experiment. Acidic normoxia showed a significant decrease by 26% ($P = 0.03$), while acidic hypoxia showed a 10% increase as compared with the nonacidic conditions. This demonstrates that the stromal lactate concentration was similar in all the corneal buttons and did not contribute a significant factor in the relative differences between the conditions.

Previous studies¹² in our laboratory showed that lactate efflux from stroma to aqueous across the endothelium was facilitated via membrane lactate transport systems including apical lactate:H⁺ cotransport. Conceivably, the low pH condition used in the current study could slow lactate efflux from the cornea by virtue of the less favorable proton gradient. To eliminate possible effects of Ringer pH on lactate efflux via this cotransporter, we repeated a series of experiments in the absence of endothelium. This was achieved by placing each cornea, epithelial side down, into an egg cup and gently scraping the endothelium off with a cotton tipped applicator. Corneas were then rinsed in the appropriate Ringer solution and placed in

individual vials containing 3 mL Ringer equilibrated under the conditions of normal and acidic pH without CO₂. If the lactate transporter was a rate limiting step, then relative changes in lactate efflux between normal and acid pH would be diminished.

Table I (trial #5) shows the results in the absence of the endothelium (no CO₂). Overall, lactate production was slightly lower when compared with trial #3 (also no CO₂). The acidic normoxic condition decreased lactate production by 33% ($P < 0.05$) versus the nonacidic normoxic condition. The acidic hypoxic condition decreased lactate production by 17% ($P = 0.035$) as compared with the nonacidic hypoxic condition. Thus, acidosis still depressed lactate production in the absence of the endothelium indicating that the possible pH effect on the lactate transporters was not apparent under these conditions.

Discussion

The focus of this study was to determine whether acidic conditions, especially during hypoxia, reduce lactate formation by the cornea. Corneal swelling studies with human subjects have shown that hypoxic corneas made more acidic by exposure to CO₂ swell less than corneas exposed to hypoxia alone.⁹ Since hypoxic swelling is coupled with lactate production in the cornea,^{1,2} it is reasonable to suggest that the lower pH could be affecting lactate production. The results presented in Table I show that acidic conditions have a strong inhibitory effect on lactate production (20%–30%) in the presence of air (acidic normoxia). However, this inhibition is substantially diminished under hypoxic conditions (acidic hypoxia).

Five experimental protocols were used to test differences between acidic and nonacidic normoxia and between acidic and nonacidic hypoxia. The first protocol used continuous gas bubbling and sampling every 10 minutes with replacement. Lactate production was significantly reduced under normoxic and hypoxic conditions using this protocol. Concerns about variable rates of evaporation under these conditions prompted us to modify the protocol (trial #2) where the pre-gassed solution was sealed in the vial and sampling was done once after 60 minutes. However, this protocol did not reduce the standard deviation of the measurements. The effect of acidosis on lactate production under normoxic conditions was significant, but the

decrease was less than in trial #1. Under hypoxic conditions, no difference in lactate production was observed. In trials #3 and #5, lactate production was measured in the absence of $\text{CO}_2/\text{HCO}_3^-$ to test if the effect on lactate production was simply due to the bath pH or sensitivity to $\text{CO}_2/\text{HCO}_3^-$, as well. Significant decreases during acidic normoxia (-25% and -33%) and a tendency to uncover decreases during acidic hypoxia (-9% and -17%) were found in the absence of $\text{CO}_2/\text{HCO}_3^-$. If the reduction in lactate production is being influenced by intracellular pH (pHi, see below), then the absence of $\text{CO}_2/\text{HCO}_3^-$ may be allowing the pHi to go lower under the acidic conditions. $\text{CO}_2/\text{HCO}_3^-$ is known to substantially increase intracellular buffering power.^{13,14} Therefore, it is likely that in its absence, pHi will be lower during the acidic conditions.

Previous studies by Chen and Chen¹¹ also demonstrated reduced lactate formation under acidic conditions. They recommended pre-incubating corneas in glucose-free Ringer to remove extracellular lactate already present in the corneas, thereby giving a purer measure of lactate production. In trial #4, where we did this pre-incubation, lactate levels were considerably lower in all four conditions, consistent with removing lactate already present in the cornea. A 26% decrease in lactate production was observed under acidic normoxic conditions, while no decrease was observed under acidic hypoxia. Chen and Chen¹¹ showed an 11% decrease in lactate production at pH 6.7 relative to pH 7.4 under normoxic conditions and a 13% decrease under hypoxic conditions. The different experimental conditions may have produced this result.

First, in our experiments, the pH differences were significantly less (7.5 versus 7.05). Second, whereas Chen and Chen used rabbit corneas, non- $\text{CO}_2/\text{HCO}_3^-$ Ringers, and antimycin A plus rotenone to produce a chemical hypoxia, we used bovine corneas and conditions ($\text{CO}_2/\text{HCO}_3^-$, nitrogen gas) designed to simulate the goggle experiments used in humans. Thus, although both studies demonstrated a significant effect due to low pH bathing solutions, we showed a definite attenuation of this effect under hypoxia while Chen and Chen¹¹ did not. Finally, we removed the endothelium to determine if endothelial lactate transporters¹² may have influenced the rate of lactate efflux from the cornea. We found reduced lactate production overall, consistent with removal of cells, but no effect on the relative changes in lactate production. A weighted average of all five trials produced a 6% reduction in lactate production under acidic hypoxia. However, when only trials where CO_2 was present were used, the drop was only 1.8%.

The mechanisms by which pH can affect glycolytic activity are well studied. The key regulatory enzyme in glycolysis, phosphofructokinase (PFK), is inhibited by reductions in pH. Changes in bath pH are generally manifest in the cytoplasm although the intracellular changes are smaller due to intrinsic buffering and pH regulatory transporters.^{14,15} Therefore, it is likely that the effect observed in our study was due to inhibition of glycolysis, consistent with previous studies.¹¹ However, PFK is acutely sensitive to the energy status of the cell, e.g., the ATP/ADP ratio.¹⁰ A low ratio significantly activates PFK. During hypoxia, this ratio will decrease. Glycolytic activity thereby

increases and more lactate is produced, as seen in Table I. At low ATP/ADP ratios, the pH sensitivity also shifts to lower pH values.¹⁰ Thus, during hypoxia, which is accompanied by decreased pHi,⁴ PFK activity is less sensitive to pH so that energy production is not hampered during this stressed period. The attenuated reduction in lactate production caused by acidosis under hypoxia that we observed is consistent with this interpretation.

The results from this pilot study indicate that lactate production may be slightly reduced during acidic hypoxia relative to hypoxia alone. Thus, reduced lactate production cannot be eliminated as a possible cause for the reduced corneal swelling observed under acidic hypoxic conditions,⁹ although the small differences make it unlikely. It is possible that this question could be definitively addressed by measuring corneal swelling and lactate production simultaneously^{1,2} under these same conditions. McNamara and coworkers⁹ also suggested that the reduced swelling under acidic conditions could be produced by a reduction in endothelial hydraulic conductivity, which may coincide with the acute and chronic changes in endothelial morphology seen during acidosis and contact lens wear.^{6,7}

References

1. Klyce S: Stromal lactate accumulation can account for corneal oedema osmotically following epithelial hypoxia in the rabbit. *J Physiol* 1981;321:49-64.
2. Rohde M, Huff J: Contact lens-induced edema in vitro-amelioration by lactate dehydrogenase inhibitors. *Curr Eye Res* 1986;5:751-758.
3. Giasson C, Bonanno JA: Corneal epithelial and aqueous humor acidification during *in vivo* contact lens wear in rabbits. *Invest Ophthalmol & Vis Sci* 1994;35:851-861.
4. Bonanno JA, Polse KA: Corneal acidosis during contact lens wear: effects of hypoxia and CO_2 . *Invest Ophthalmol & Vis Sci* 1987;28:1514-1520.
5. Giasson C, Bonanno JA: Acidification of rabbit corneal endothelium during contact lens wear in vitro. *Curr Eye Res* 1995;14:311-318.
6. Holden B, Williams L, Zantos S: The etiology of transient endothelial changes in the human cornea. *Invest Ophthalmol & Vis Sci* 1985;26:1354-1359.
7. MacRae S, Matsuda M, Shellans S, et al: The effects of hard and soft contact lenses on the corneal endothelium. *Am J Ophthalmol* 1986;102:50-57.
8. Polse K, Brand R, Cohen S, et al: Hypoxic effects on corneal morphology and function. *Invest Ophthalmol & Vis Sci* 1990;31:1542-1554.
9. McNamara NA, Polse KA, Bonanno JA: Stromal acidosis modulates corneal swelling. *Invest Ophthalmol & Vis Sci* 1994;35:846-850.
10. Trivedi B, Danforth W: Effect of pH on the kinetics of frog muscle phosphofructokinase. *J Biol Chem* 1966;241:4110-4114.
11. Chen C-H, Chen S: Lactate transport and glycolytic activity in the freshly isolate rabbit cornea. *Arch Biochem Biophys* 1990;276:70-76.
12. Giasson C, Bonanno JA: Facilitated transport of lactate by rabbit corneal endothelium. *Exp Eye Res* 1994;59:73-81.
13. Bonanno JA, Giasson C: Intracellular pH regulation in fresh and cultured bovine corneal endothelium. I. Na/H exchange in the absence and presence of HCO_3^- . *Invest Ophthalmol & Vis Sci* 1992;33:3058-3067.
14. Roos A, Boron W: Intracellular pH. *Physiological Reviews* 1981;61:297-432.
15. Bonanno JA, Machen TE: Intracellular pH regulation in basal corneal epithelial cells measured in corneal explants: characterization of Na/H exchange. *Exp Eye Res* 1989;49:129-142.

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